BL2028N Datasheet

802.11n + Bluetooth 5.2

Beken Corporation
Building 41, 1387 Zhangdong Rd, Shanghai 201203, China
PHONE: (86) 21 5108 6811
FAX: (86) 21 6087 1089

Contents

Revision History

Version	Description	Date
1.0	Initial release.	Mar/17/2021

1. Introduction

BL2028N is a highly integrated dual-mode Bluetooth 5.2 and Wi-Fi 802.11n combo solution with complete hardware and software resources needed for Wi-Fi and Bluetooth applications. It integrates Bluetooth Classic i.e. Basic Rate (BR) and Enhanced Data Rate (EDR) as well as Bluetooth Low Energy (LE) features fully compliant with the Bluetooth 5.2 specification. It supports AP and STA mode concurrently.

A 32-bit MCU running up to 120 MHz and built-in 256 KB RAM enable the chip to support multi-cloud connectivity. The MCU's extended instructions specifically for signal processing allow for efficient audio encoding and decoding.

BL2028N includes a rich set of peripherals, such as PWM, I2C, UART for program download and burning, SPI, SDIO and IrDA. BL2028N can provide up to six 32-bit high-speed PWM channels for high-quality LED control. Each of the two PWM channels can be configured as a phase-controlled differential mode to support motor and strip drive.

BL2028N integrates a Packet Traffic Arbitration (PTA) interface to facilitate coexistence of Bluetooth and 802.11 WLAN.

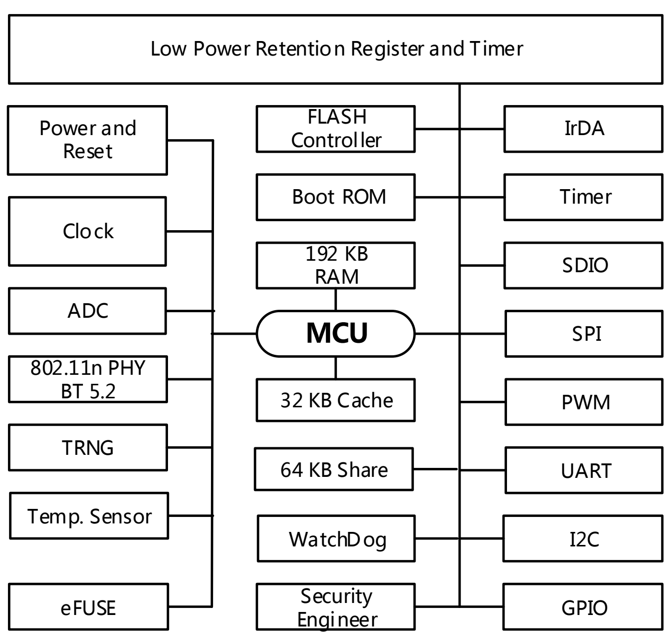
BL2028N embeds eFUSE and supports OTP read/write in FLASH, which can be used to provide unique ID, code encryption and secure the debug interface. A built-in True Random Number Generator (TRNG) and a security module are integrated to ensure secure communication and fast authentication and network connectivity.

BL2028N supports low power sleep modes where the MCU can enters sleep modes with a micro amp level. In deep sleep mode, the chip can run a 32-bit clock with a few microamperes of current and can be woken up by this clock or by any GPIO.

BL2028N supports Bluetooth Classic i.e. Basic Rate (BR) and Enhanced Data Rate (EDR) as well as all Bluetooth LE 5.2 rates and features, including Long Range, High Data Rate, and angle-of-arrival (AoA) and angle-of-departure (AoD) positioning with up to four antennas.

2. Features

- 802.11 b/g/n 1x1 compliant
- 17 dBm TX output power at 54 Mbps
- -76 dBm RX Sensitivity at 54 Mbps
- 20 MHz bandwidth and STBC
- Working mode STA, AP, Direct
- Concurrent AP + STA
- MCU at up to 120 MHz
- 256 KB internal data RAM
- 2 MB or 4 MB Flash
- UART or SPI download
- SDIO up to 50 MHz clock rate
- High-speed SPI
- 2 x I2C interface
- 6 x 32-bit timer and 1 x low-power timer
- 6 x PWM channels with either high-speed clock or low-power clock
- 6 x 10-bit ADC with internal decimation filter up to 16 bits
- 32-byte eFUSE
- 256B ~ 2 KB OTP
- True Random Number Generator (TRNG)
- 26 MHz and 32 kHz clock signal output
- Power-on reset (POR) and watchdog reset



3. Pin description

BL2028N provides Wi-Fi and Bluetooth functionality in a 4x4 mm, 32-pin QFN package. The figure below shows the pin definition of BL2028N.

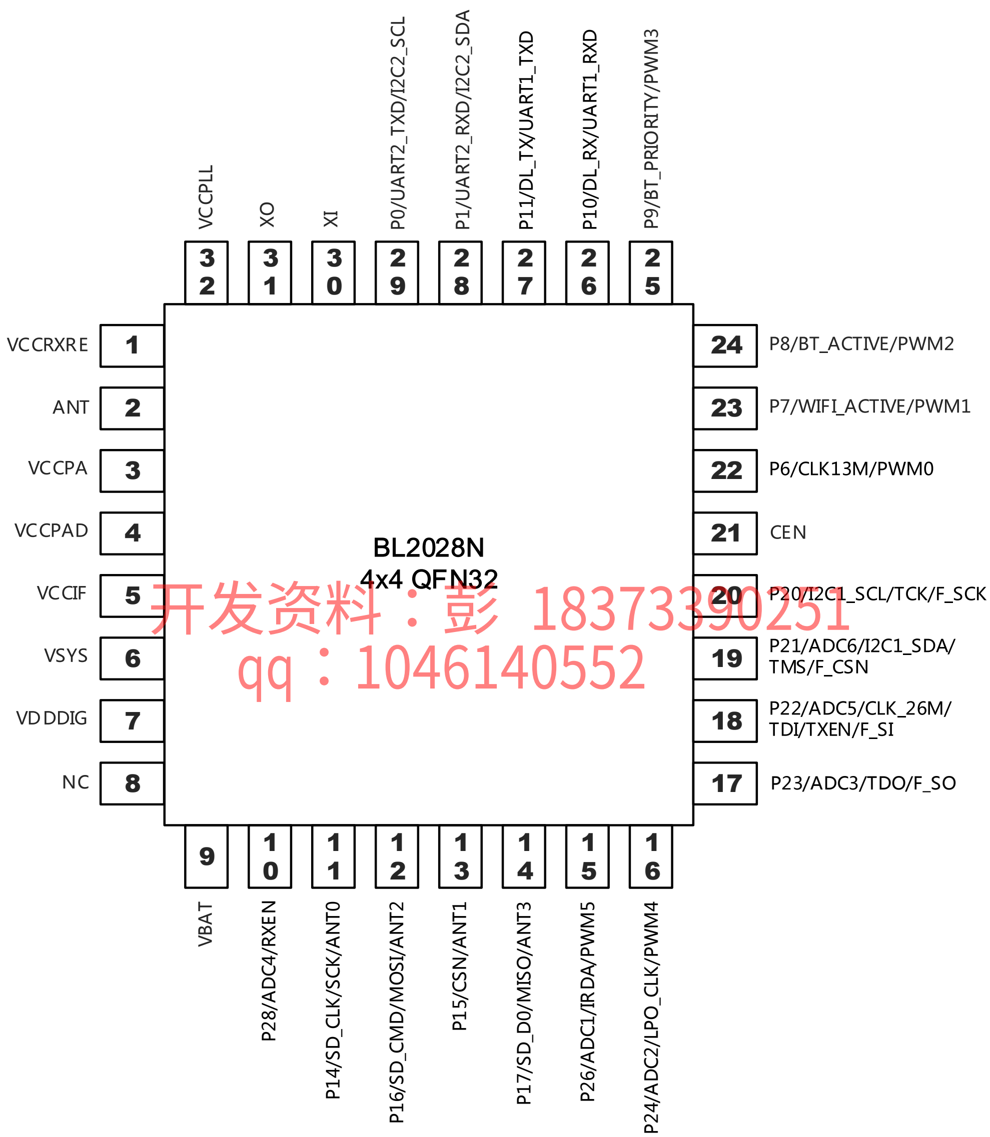


Figure 1 BL2028N 32-Pin Definition

Table 1 BL2028N Pin Descriptions

Pin #	Pin Name	I/O Type	Description
1	VCCRXFE	I	RF receiver power supply
2	ANT	IO	2.4 GHz RF signal port
3	VCCPA	I	RF PA power supply
4	VCCPAD	I	RF transmitter power supply
5	VCCIF	I	IF power supply
6	VSYS	O	System LDO output
7	VDDDIG	O	Digital LDO output, ~1.2 V
8	NC	NC	No connect
9	VBAT	I	Chip power supply
10	P28/ADC4/RXEN	IO	GPIO or ADC4 or RX enable
11	P14/SD_CLK/SCK/ANT0	IO	GPIO or SD Card Clock or SPI Clock or BLE antenna select ANT0
12	P16/SD_CMD/MOSI/ANT2	IO	GPIO or SD Card CMD or SPI MOSI or BLE antenna select ANT2
13	P15/CSN/ANT1	IO	GPIO, SPI CSN or BLE antenna select ANT1
14	P17/SD_D0/MISO/ANT3	IO	GPIO or SD Card DATA0 or SPI MISO or BLE antenna select ANT3
15	P26/ADC1/IRDA/PWM5	IO	GPIO or ADC1 or IrDA input or PWM5
16	P24/ADC2/LPO_CLK/PWM4	IO	GPIO or ADC2 or low-power 32.768 kHz clock output or PWM 4
17	P23/ADC3/TDO/F_SO	IO	GPIO or ADC3 or JTAG TDO or SPI Flash data output F_SO
18	P22/ADC5/CLK_26M/TDI/TXEN/F_SI	IO	GPIO or ADC5 or 26 MHz clock output or JTAG TDI or TX enable or SPI Flash data input F_SI
19	P21/ADC6/I2C1_SDA/TMS/F_CSN	IO	GPIO or ADC6 or I2C1 SDA or JTAG TMS or SPI Flash CSN
20	P20/I2C1_SCL/TCK/F_SCK	IO	GPIO or I2C1 SCL or JTAG TCK or SPI

Pin #	Pin Name	I/O Type	Description
			Flash clock
21	CEN	I	Chip enable, active high
22	P6/CLK13M/PWM0	IO	GPIO or 13 MHz clock output (X26M divided by 1/2/4/8) or PWM0SS
23	P7/WIFI_ACTIVE/PWM1	IO	GPIO or PTA WIFI_ACTIVE or PWM1
24	P8/BT_ACTIVE/PWM2	IO	GPIO or PTA BT_ACTIVE or PWM2
25	P9/BT_PRIORITY/PWM3	IO	GPIO or PTA BT_PRIORITY or PWM3
26	P10/DL_RX/UART1_RXD	IO	GPIO or UART RXD or UART1 RXD
27	P11/DL_TX/UART1_TXD	IO	GPIO or UART TXD or UART1 TXD
28	P1/UART2_RXD/I2C2_SDA	IO	GPIO or UART2 RXD or I2C2 SDA
29	P0/UART2_TXD/I2C2_SCL	IO	GPIO or UART2 TXD or I2C2 SCL
30	XI	I	26 MHz Crystal input
31	XO	O	26 MHz Crystal output
32	VCCPLL	I	RF PLL power supply

4. Wi-Fi and Bluetooth

BL2028N supports full features of 802.11b/g/n. With the transmit activation indicator TXEN of GPIO22 and the receive activation indicator RXEN of GPIO28, the system can be externally connected to LNA and PA to achieve longer communication distance.

BL2028N integrates Bluetooth classic and low energy BLE system, Bluetooth and Wi-Fi share antenna and transceiver circuits. The built-in packet traffic arbitration interface ensures stable Bluetooth and Wi-Fi dual connectivity and enables efficient sharing of over-the-air resources.

5. Clock

The system has six root clock signals: X26M, DCO, X32K, D32K, ROSC, and DPLL.

- X26M: High frequency crystal oscillator. X26M, typically running at 26 MHz, is used for MCU and other peripherals and is also the reference clock signal for digital PLL (DPLL) and audio PLL (I2SPLL). X26M has a tunable load capacitance from 6 to 18 pF on both input and output ports with 64 steps, therefore, no external capacitance is needed. The startup time of X26M is about one milliseconds.
- DCO: Internal high frequency digital controlled oscillator running at 26 MHz to 120 MHz with+/-2% frequency tolerance after calibration. The startup time of DCO is about a few microseconds.
- X32K: High frequency crystal oscillator, typical frequency 32.768 kHz
- D32K: 32 kHz clock signal divided from X26M
- ROSC: Internal low frequency ring oscillator with ±2% frequency tolerance after calibration

- DPLL: High speed 480 MHz PLL clock

Low power clock (LPO_CLK) is selected from X32K, D32K or ROSC.

The MCU and peripheral clock selection options are listed as follows.

Table 2 Clock Selection

MCU and Peripherals	X26M	DCO	DPLL	LPO_CLK
MCU	√	√	√	√
ADC	√	√		
SDIO	√	√		
PWM	√	√		√
SPI	√	√	√	
I2C2	√	√		
IrDA	√	√		
I2C1	√	√		
UART2	√	√		

MCU and Peripherals	X26M	DCO	DPLL	LPO_CLK
UART1	√	√		
Timer 1	√			
Timer 2				√
Always on (AON) timer				√

The system can output clock signal for external components.

- LPO_CLK and X26M clock can be output to GPIOs for general purpose.
- 13 MHz clock derived from X26M provides clock for FM receiver

6. Reset

There are multiple sources that can trigger a reset, including always on (AON) logic reset, system power-on reset, digital power-on reset and watchdog reset. Any reset will reset the whole chip to initial state. Watchdog resets have special indication bits to distinguish them from power-on resets. The AON logic has one 32-bit timer and a 16-bit retention register. The 32-bit timers could only be reset to initial value by system power on reset.

Waking up from either shutdown mode or deep sleep mode will power on digital from power down mode, that trigger the whole system reset procedure.

7. Power management

The Power Management (PM) module in BL2028N is designed for ultra-low power applications. To reduce power consumption, BL2028N can enter the four low power modes as follows.

Shutdown Mode – In this mode all circuits are powered down when CEN = 0, and the system will power on only after CEN back and keep high for a few milliseconds.

Deep Sleep Mode – In this mode all circuits are powered down except GPIO and always on logic. Any GPIO edge transition or AON timer timeout event can power up the system again. The retention register could keep its contents at this mode.

Normal Standby Mode – In this mode the MCU stops running and all peripheral interrupt can resume MCU.

Low Voltage Standby Mode – In this mode the MCU and all digital logic stop their clocks, and the power supply of them decreases to a much lower retention voltage, thus the current could be much lower. In this mode, only GPIO and RTC timer could resume the system to run mode with normal voltage.

8. Peripherals

8.1. UART

BL2028N includes two Universal Asynchronous Receiver/Transmitter (UART) interfaces, which offer full-duplex, asynchronous serial communication at a baud rate up to 6 Mbps. It supports 5/6/7/8 bits data, and even, odd or none parity check. The stop bit can be either 1 bit or 2 bits. UART1 (GPIO10=DL_RX，GPIO11=DL_TX) supports Flash download.

In low voltage standby mode, low level continuously applying to TXD or RTD will wake up the MCU and the UART and MCU can resume to active mode.

8.2. SPI

BL2028N integrates a SPI interface. The SPI provides an interface for giving instructions to BL2028N and data communication. The SPI 4-wire master/slave interface consists of three input signals (SCK, CSN, and MOSI) and one output signal (MISO).

BL2028N supports a high-speed SPI slave interface with maximum 50 MHz clock speed,. This high speed SPI has a dedicated DMA channel that could work at high speed without MCU load.

BL2028N also supports a low-speed SPI master/slave interface with up to 8 MHz clock speed,. The receive data could be latched on either rising edge or falling edge of clock signal. The transmit data could be set by MSB or LSB first.

8.3. SDIO

BL2028N SDIO has master mode. The SDIO supports a maximum 50 MHz clock speed with one bit to four bits. The SDIO could be used to read external SD card as master mode.

The SDIO has a dedicated DMA channel that could work at high speed without MCU load.

8.4. I2C

I2C is a popular inter-IC interface that requires only two bus lines, the serial data line (SDA) and the serial clock line (SCL). BL2028N embeds two I2C hardware interfaces, which could act as Master mode or Slave mode. The I2C supports a normal 400 kHz clock speed with 7 bit addressing. If low level on SCL or bus idle duration is greater than a programmable threshold, it will generate interrupt to MCU.

8.5. ADC

BL2028N includes a 10~16 bits ADC module with internal decimation filter, supporting up to six external analog input channels. The ADC is a differential successive approximation register (SAR) analog-to-digital converter, which can be used to sample analog input signals such as GPIO and temperature sensor. It can be operated in a one-shot mode with sampling under software control, or a continuous mode with a programmable sampling rate, or a software read mode.

Besides GPIOs, the system power VSYS, temperature sensor and internal calibration circuit can also serve as the input channels for the ADC.

8.6. PWM

BL2028N has six Timers (no PWM) and six 32-bit PWM channels (Timer mode supported). Each PWM channel has three modes: Timer mode, PWM mode, and Capture mode. Each mode of each channel is multiplexed with 32-bit counting. The PWM running clock can be either high speed clock or low power clock. Each PWM runs independently with its own duty cycle.

The main features of the PWM module are listed here:

- Fixed PWM base frequency with programmable 1 ~ 256 MHz prescaler
- The counter increases in one direction and continues counting from 0 automatically when it overflows to the maximum value.
- Each channel can be individually enabled, and the mode of each channel can be individually configured.
- Capable of continuous counting between two rising edges, two falling edges or dual edges in Capture mode
- Configurable PWM period and duty cycle for each PWM channel
- Real-time count value can be read in Timer mode.

For BL2028N, currently only the first group of PWM0 ~ PWM5 signals are mapped to GPIOs and the second group is only used as Timer.

8.6.1. Timer mode

In Timer mode, the counter is enabled and incrementally counted, and an interrupt is generated when the specified cycle value is reached. Counting restarts from 0.

If the software refreshes the count cycle value during the counting process, the new count cycle value is used as the count cycle. If the current count value exceeds the new count cycle value, it immediately goes back to 0 and starts counting again.

The counter resets to 0 immediately if Enable = 0. If Stop = 1, the counter stops incrementing and remains its current state, and continues counting after Stop = 0.

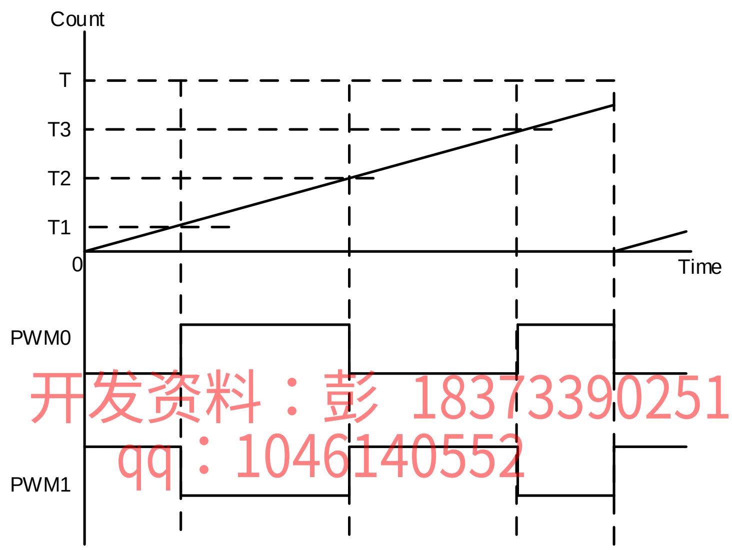
The current count value of the counter can be read in real time.

8.6.2. PWM mode

In PWM mode, the PWM waveform start level can be configured as 0 or 1. The waveform timing is configured with four parameters.

- Waveform period T (1~2^32)
- The first level inversion time T1 (0~2^32-1, 0 - no inversion)
- The second level inversion time T2 (0~2^32-1, 0 - no inversion)
- The third level inversion time T3 (0~2^32-1, 0 - no inversion)

Each group of six PWM channels can be configured as three pairs, and their start up time is aligned when the adjacent two channels are in paired mode (At this point, the waveform period must be configured to the same value).



As shown in the above figure, PWM0 and PWM1 have opposite start levels, share same waveform parameters, and are in paired mode.

In PWM mode, no interrupts are generated.

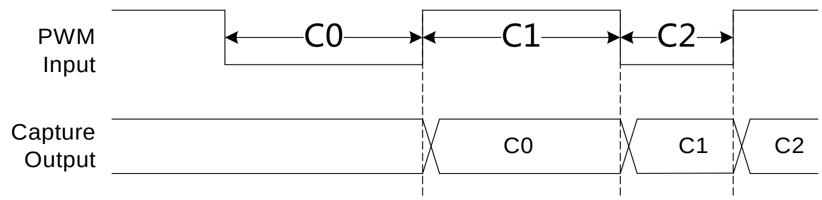
During operation, any updates to the configuration parameters will take no effect until the next time the counter starts from 0 again.

8.6.3. Capture mode

Capture mode, which uses the operating clock to count the time between input signal edges, has the following three modes of edges:

- Rising edge: between two rising edges
- Falling edge: between two falling edges

- Dual edge: between any two edges



The figure above shows the Capture result between dual edges.

Each time there is an update to the Capture output, an interrupt is generated. The software must read out the Capture result before the next update, otherwise the Capture result will be overwritten by the new result.

8.7. Timer

BL2028N has two groups of peripheral timers; each group has three 32-bit timers. The first group runs with crystal clock X26M with a 4-bit prescaler. The second group runs with a low-power clock with another 4-bit prescaler.

The watchdog timer is used to reset the system to offer robust protection against system disorder. The watchdog also stops when the MCU stops running or when power is lost.

The AON timer is used for low-power timing and it can keep running even when the MCU is powered off.

8.8. GPIO

BL2028N has total 19 GPIOs. Any GPIOs can be configured as an interrupt source to interrupt system at active mode or wake up system from sleep mode.

The GPIO has peripheral function as the table below.

Table 3 GPIO Peripheral Function

Pin #	GPIO	Peripheral Function
10	GPIO28	ADC4/RXEN
11	GPIO14	SD_CLK/SCK/ANT0
12	GPIO16	SD_CMD/MOSI/ANT2

Pin #	GPIO	Peripheral Function
13	GPIO15	CSN/ANT1
14	GPIO17	SD_D0/MISO/ANT3
15	GPIO26	ADC1/IRDA/PWM5
16	GPIO24	ADC2/LPO_CLK/PWM4
17	GPIO23	ADC3/TDO/F_SO
18	GPIO22	ADC5/CLK_26M/TDI/TXEN/F_SI
19	GPIO21	ADC6/I2C1_SDA/TMS/F_CSN
20	GPIO20	I2C1_SCL/TCK/F_SCK
22	GPIO6	CLK13M/PWM0
23	GPIO7	WIFI_ACTIVE/PWM1
24	GPIO8	BT_ACTIVE/PWM2
25	GPIO9	BT_PRIORITY/PWM3
26	GPIO10	DL_RX/UART1_RXD
27	GPIO11	DL_TX/UART1_TXD
28	GPIO1	UART2_RXD/I2C2_SDA
29	GPIO0	UART2_TXD/I2C2_SCL

8.9. Flash download

Method 1: Download through UART (GPIO10=DL_RX，GPIO11=DL_TX)

Method 2: Download through dedicated SPI. Any time the digital logic resets to active, within a few hundred milliseconds the GPIO20 ~ GPIO23 will act as mode selection port and then back to normal GPIO mode. If there is FLASH download command, these four ports will change to FLASH download port that FLASH could be programming.

8.10. IrDA interface

BL2028N includes a hardware IrDA decoder interface to decode the signal which supports NEC. In addition, the interface has the capture timer capability to allow software decoding of the input signal.

8.11. Security

BL2028N ensures system and communication security by incorporating True Random Number Generator (TRNG), eFUSE, OTP and hardware signature and verification.

The 32-byte eFUSE could be written through download port GPIO20 ~ GPIO23 at download mode. The eFUSE has defined function mapping as follows. All bytes could be read by MCU and download port except low 16 bytes could not be read out by download port.

Table 4 eFUSE Definition

Byte	Description
Byte 31	bit 7 : 1 - disable JTAG; 0 - enable JTAG bit 6 : 1 - disable FLASH download; 0 - enable FLASH download bit 5 : 1 - byte 15:0 for code encryption; 0 - byte 15:0 for user use, no code encryption bit 4: 1 – byte 15:0 read disable bit 3: 1 - byte 15:0 write disable bit 2: 1 - byte 16:23 write disable bit 1: 1 - byte 24:29 write disable bit 0: 1 - eFUSE byte 32 write disable **Note:** • The above read disable operation is only valid when byte 15:0 is used for code encryption; otherwise byte 15:0 is always readable. • The above write disable operation applies to burning ports and MCU.
Byte 30:16	User Defined such as MAC address
Byte 15:0	Code Encryption

8.12. Temperature sensor

BL2028N integrates an on-chip temperature sensor. The temperature sensor can measure on-chip temperature over -40 to +125℃ with a precision of +/-3℃. The digital results can be read by the ADC.

Usually the software initiates calibration of a specific module based on the temperature value, narrowing the difference in chip performance at different temperatures. The host can also read the on-chip temperature and decide whether to reduce the transmit power or suspend operation at high temperatures.

9. Electrical characteristics

9.1. Absolute maximum ratings

Item	Pin Name	Min.	Max.	Unit
Battery and Power Supply IO	VCCRXFE, VCCPA, VCCPAD, VCCIF, VSYS, VBAT, CEN, VCCPLL	-0.3	3.9	V
Digital Input Pin		-0.3	3.9	V
Analog Pin	XI, XO	-0.3	1.5	
RF Pin	ANT	-0.3	1.5	V
Storage Temperature		-55	125	℃

9.2. ESD ratings

Item	Description	Value	Unit
ESD_HBM	Electrostatic Discharge Tolerance under Human Body Model	+/-4000	V
ESD_MM	Electrostatic Discharge Tolerance under Man Machine Model	+/-200	V
ESD_CDM	Electrostatic Discharge Tolerance under Charged Device Model	+/-250	V

9.3. Recommended operating conditions

Parameter	Condition	Min.	Typ.	Max.	Unit
Operating supply voltage	VBAT-pin	2.1	3.3	3.6	V
Ambient operating		-40		105	℃

Parameter	Condition	Min.	Typ.	Max.	Unit
temperature					
High-level output voltage V_{OH}	I_{OH} = -0.25 mA	VBAT-0.3		VBAT	V
Low-level output voltage V_{OL}	I_{OH} = 0.25 mA	VSS		VSS+0.3	V
IO drive strength	Configured through registers		6	20	mA

9.4. Power consumption

Parameter	Condition	Min.	Typ.	Max.	Unit
TX current	14 dBm, HT20 MCS7		250		mA
	15 dBm, 54 Mbps OFDM		250		mA
	17 dBm, 11 Mbps DSSS		280		mA
RX current	-10 dBm input, 54 Mbps OFDM		69		mA
	-10 dBm input, HT20 MCS7		69		mA
	-10 dBm input, 11 Mbps DSSS		63		mA
Normal standby current	MCU stops running, Modem is powered off, and the system could be woke up by internal timer		30		uA
Low voltage standby current	MCU stops running and enters low voltage and the system could only be woke up by GPIO or RTC timer		10		uA
Deep sleep current	All blocks are powered off except the AON timer.		5		uA
Shutdown current	CEN=0		1		uA
Note: All measurements are made under the following conditions: Ambient temperature = 25 °C; VDD = 3.3 V, duty cycle = 90%, continuous receive mode					

9.5. WLAN receiver characteristics

Parameter		Condition	Min.	Typ.	Max.	Unit
Operating frequency			2412		2484	MHz
Sensitivity (20 MHz)	802.11n	MCS0		-93		dBm
		MCS1		-90		dBm
		MCS2		-88		dBm
		MCS3		-85		dBm
		MCS4		-82		dBm
		MCS5		-78		dBm
		MCS6		-77		dBm
		MCS7		-75		dBm
	802.11g	6 Mbps		-95		dBm
		9 Mbps		-92		dBm
		12 Mbps		-91		dBm
		18 Mbps		-89		dBm
		24 Mbps		-86		dBm
		36 Mbps		-83		dBm
		48 Mbps		-78		dBm
		54 Mbps		-77		dBm
	802.11b	1 Mbps		-99		dBm
		2 Mbps		-95		dBm
		5.5 Mbps		-93		dBm
		11 Mbps		-91		dBm
Adjacent Channel interference C/I		HT20 MCS7		25		dB
		54 Mbps OFDM		26		dB
		11 Mbps DSSS		40		dB

9.6. WLAN transmitter characteristics

Parameter		Condition	Min.	Typ.	Max.	Unit
Operating frequency			2412		2484	MHz
TX Power (EVM compliant)	802.11n	MCS7		16		dBm
		MCS0		18		dBm
	802.11g	54 Mbps		17		dBm
		6 Mbps		18		dBm
	802.11b	1/11 Mbps		19		dBm

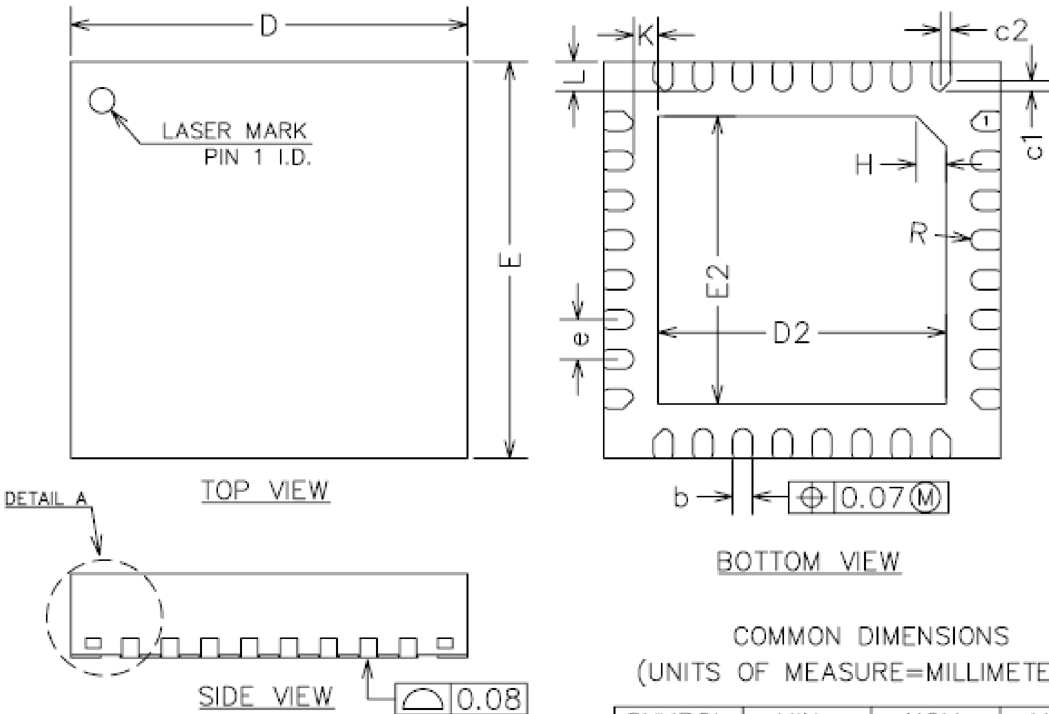
9.7. BLE receiver characteristics

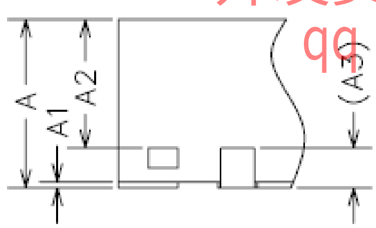
Parameter	Condition	Min.	Typ.	Max.	Unit
Operating frequency		2402		2480	MHz
On-the-air data rate			1		Mbps
Sensitivity			-97		dBm
Maximum RF signal input		-10			dBm
Intermodulation				-23	dBm
Co-Channel interference $C/I_{co\text{-}channel}$			10		dB
Adjacent channel interference C/I	+1MHz		0		dB
	-1MHz		0		dB
	+2MHz		-20		dB
	-2MHz		-27		dB
	+3MHz		-25		dB
	-3MHz		-36		dB
Out-of-band blocking	30 MHz ~2000 MHz	-10			dB
	2000 MHz ~2400 MHz	-20			dB
	2500 MHz ~3000 MHz	-10			dB
	3000 MHz ~12.5 GHz	-10			dB

9.8. BLE transmitter characteristics

Parameter	Condition	Min.	Typ.	Max.	Unit
Operating frequency		2402		2480	MHz
On-the-air data rate			1		Mbps
TX power		-20	5	20	dBm
20 dB bandwidth			1		MHz
Frequency offset		-150		150	KHz
Max drift		-50		50	KHz
Drift rate			80	400	Hz/us
△f1avg		225	244	275	KHz
△f2max		185	195		KHz
△f1avg/△f2avg		0.8	0.85		
Adjacent channel TX power	2 MHz offset		-45	-20	dBm
	>= 3 MHz offset		-47	-30	dBm

10. QFN32 4 X 4 mm package





DETAIL A

COMMON DIMENSIONS
(UNITS OF MEASURE=MILLIMETER)

SYMBOL	MIN	NOM	MAX
A	0.80	0.85	0.90
A1	0	0.02	0.05
A2	0.60	0.65	0.70
A3		0.20REF	
b	0.15	0.20	0.25
D	3.90	4.00	4.10
E	3.90	4.00	4.10
D2	2.80	2.90	3.00
E2	2.80	2.90	3.00
e	0.30	0.40	0.50
H		0.30REF	
K		0.25REF	
L	0.25	0.30	0.35
R	0.09	–	–
c1	–	0.10	–
c2	–	0.10	–

11. Ordering information

Part Number	Package	Packing	MPQ(ea)
BL2028NQN32	QFN32_4X4	Tape Reel	3K